CARVEN MACEKE

Junior Software Engineer | Java, Spring Boot, React, SQL

Address: Pretoria West, Pretoria, South Africa
Cell Number: 060 795 0837 | Email: carvenmaceke9@gmail.com
Portfolio: carven.axeconnect.co.za | GitHub: github.com/Carvenmaceke | LinkedIn: linkedin.com/in/carven-maceke-a29985235

PROFILE

Junior Software Engineer with a National Diploma in Information Technology (Software Development) who looks for the slow, manual and error-prone parts of daily work and builds software that removes them. At K & K Media I noticed that payslips, leave requests and IT issues were all handled by hand, so on my own initiative I designed and built an HR and Office Operations Platform that now runs the company's employee records, payroll, payslips, leave, IT support and office issue reporting in production. I work daily with Java 17, Spring Boot 3, Representational State Transfer (REST) APIs, MySQL and React, and I am currently a Software Development Intern at Forge Academy x Nokia. AWS Certified Cloud Practitioner, seeking a Junior Software Engineer, Backend or Full-Stack Developer role.

KEY COMPETENCIES

- Java and Spring Boot Development
- Full-Stack Web Development (React)
- Authentication and Security (JWT, RBAC)
- CI/CD and Cloud Deployment
- Requirements Gathering with End Users
- REST API Design and Integration
- SQL and Relational Database Design
- Workflow and Process Automation
- Problem Solving and Troubleshooting
- Mentoring and Team Collaboration

WORK EXPERIENCE

K & K Media & Solwazi Institute | Junior IT Specialist **May 2025 – Present**
Midrand, South Africa

- Spotted that payslips, leave requests and IT problems were all handled manually, and took the initiative, alongside my IT support duties, to build one system that replaced all three.
- Designed, built and deployed the company's HR and Office Operations Platform from requirements to production, with a Java 17 and Spring Boot 3.3.4 REST API, a MySQL database and a React (Vite) frontend.
- Automated monthly payroll: HR captures salary, tax and banking details once, and the system calculates PAYE and UIF-style deductions and leave balances from live data instead of manual spreadsheets.
- Replaced manual payslip preparation with a five-stage approval workflow (Draft, Reviewed, Approved, Finalized, Sent) that generates branded PDF payslips with Apache PDFBox, adds a ZXing QR code for authenticity checks, and emails them through the Resend API.
- Digitised leave management with online applications, manager approvals, live leave balances and auto-generated PDF leave letters and onboarding documents.
- Added IT support ticketing and office issue reporting so staff log problems in one place and I can track and resolve them instead of relying on verbal requests.
- Secured sensitive employee and payroll data with Spring Security, JSON Web Token (JWT) authentication and role-based access control across six permission levels (Master, Admin, HR, Manager, IT Support, Employee), each with its own dashboard.
- Set up Continuous Integration/Continuous Deployment (CI/CD) with GitHub Actions and deployed to Render (API), Aiven (managed MySQL) and GitHub Pages/Netlify (frontend); moved email from SMTP to the Resend HTTPS API when the host blocked SMTP ports.
- Maintain and support the system in production, including debugging Java memory and performance issues.
- Provide IT support: troubleshoot hardware, software and Microsoft Office issues, configure email, networks and printers, maintain the WordPress website, and clean messy Excel datasets for reporting.
- Interview candidates and mentor interns through onboarding and technical development.

Forge Academy x Nokia | Software Development Intern **Mar 2026 – Present**
Roodepoort, South Africa

- Develop backend services with Spring Boot, Spring MVC, Spring Data JPA and Hibernate, exposing REST APIs used by web and IoT clients in a large internal system that combines web development, AI/ML and IoT.
- Build frontend components and data visualisation dashboards with JavaScript, TypeScript and Bootstrap.
- Integrate ESP32 IoT devices over MQTT for real-time sensor monitoring, and apply Python (Scikit-learn, OpenCV, PyTorch/TensorFlow) for predictive modelling and computer vision features.

- Work with MySQL and PostgreSQL, Docker, Kubernetes, Git/GitHub and CI/CD pipelines with automated tests across development, testing and deployment.
- Completed structured Amazon Web Services (AWS) training and earned the AWS Certified Cloud Practitioner certification during the programme.

ZAIO Institute of Technology | Junior Java Developer **Sep 2023 – Aug 2024**
Sandton, South Africa

- Developed backend services and RESTful APIs using Java and Spring Boot.
- Designed and optimised SQL queries for data processing and reporting.
- Debugged and resolved application issues in a team, and assisted with testing, deployment and configuration.
- Used Git/GitHub for version control on shared codebases and wrote technical documentation for maintainability.

PROJECTS

HR and Office Operations Platform | Production system, self-initiated at K & K Media
Java 17, Spring Boot 3.3.4, Spring Security, JWT, MySQL, React, Apache PDFBox, ZXing, GitHub Actions | github.com/Carvenmaceke/kandkmedia

- One platform for employee records and onboarding, payroll, payslips, leave, IT support tickets and office issue reports, with every module reading from the same MySQL database.
- Layered backend (controllers, services, repositories, DTOs) with route and method-level authorisation, and a public QR verification endpoint so anyone can confirm a payslip is genuine.

Portfolio Website with AI Assistant | Personal project, live
HTML, CSS, JavaScript, Claude API, prompt engineering, Netlify | carven.axeconnect.co.za

- Built a responsive portfolio with an interactive skills explorer and a conversational AI assistant, grounded in my CV data through prompt engineering, that answers recruiter questions.

Healthcare System Web Application | Academic final-year group project, Nelson Mandela University
C#, ASP.NET MVC, PHP, HTML, CSS, JavaScript, Bootstrap, Microsoft SQL Server, Git | Mar 2023 – Nov 2023

- Built a role-based clinic appointment and patient management web application as part of a development group, using the MVC pattern with an HTML, CSS and JavaScript frontend and a PHP login module.
- Followed the Software Development Life Cycle (SDLC) stages from requirements and design through development, testing and deployment, and presented the finished system.
- Designed the Microsoft SQL Server schema and CRUD operations, and used Extract, Transform, Load (ETL) processes to clean and load data into the database.

EDUCATION & CERTIFICATIONS

National Diploma in Information Technology (Software Development), NQF Level 6
Nelson Mandela University, Gqeberha | 2024

- AWS Certified Cloud Practitioner, Amazon Web Services (2026)
- Java Certification, ZAIO Institute of Technology (Dec 2023)
- SQL Certification, ZAIO Institute of Technology (Dec 2023)
- CCNAv7: Introduction to Networks, Cisco Networking Academy
- Introduction to Cybersecurity, Cisco Networking Academy
- Cybersecurity for Everyone, University of Maryland (Feb 2024)
- Quality Work Experience, YES4Youth (Sep 2024)

TECHNICAL SKILLS

Languages: Java, SQL, JavaScript, TypeScript, Python, C#, PHP, HTML5, CSS3
Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Maven, ASP.NET MVC
Frontend: React, Vite, Bootstrap, responsive web design, data visualisation
Databases: MySQL, PostgreSQL, Microsoft SQL Server, relational database design, query optimisation, ETL
Cloud & DevOps: AWS (EC2, S3, IAM, VPC, CloudWatch), Docker, Kubernetes, GitHub Actions, CI/CD, Render, Aiven, Netlify, Linux
Tools & Practices: SDLC, Git, GitHub, debugging, testing, technical documentation, MVC and layered architecture, role-based access control, AI-assisted development (Claude, ChatGPT)

REFERENCES

Available on request.